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Gene Expression Levels Found in Cervical Cancer Through Computational Analysis

Introduction

Cervical cancer (CC) is a type of cancer that grows in the lining of the cervix and moves towards the lower part of the uterus. According to the American Cancer Society, CC affects around 14,000 women in the United States. The anatomy of the cervix has two main parts, which are the endocervix and the exocervix. The endocervix is often covered with glandular cells, and it leads into the uterus and is the opening of the cervix. On the other hand, the exocervix is the outer part of the cervix and is typically covered with squamous cells. It is important to note that squamous and glandular cells are a type of epithelial cell. The cause of CC is linked with smoking, and persistent infections of the Human Papillomavirus (HPV) are linked to CC (Kiruba et al., 2025). According to the study titled "Implications of Single-Cell RNA Sequencing in Cervical Cancer: Unravelling the Molecular Landscape" by Kiruba et al., they found that by conducting single-cell RNA Sequencing (scRNA-seq), one can profile gene expression to contribute to prognostic biological markers for potential treatment strategies. Bulk RNA sequencing measures the average gene expression levels within an entire tissue sample. In comparison to the bulk RNA-seq, scRNA-seq can provide a better and more specific analysis of what types of cells are linked with CC. My main goal was to characterize the gene expression levels from a 10x Genomics dataset across multiple clusters by creating volcano plots, training neural networks, creating confusion matrices, and PCA plots. Through computational analysis, I can confirm the types of genes that are prevalent within cervical cancer to identify their roles in the disease and the immune system.

Methods

In this project, I used the 10x Genomics lab to look at a dataset titled "Human Cervical Cancer FFPE Single Cell Gene Expression Flex (Next GEM)" to utilize in scRNA-seq analysis. The data came from one human patient who had cervical cancer, and within the dataset, there were about 9,503 cells that were analyzed. When the data was opened in the CLOUPE Browser, a UMAP projection (Fig. 1) was created and illustrated about 14 different clusters. I downloaded each cluster's Log2FoldChange (L2FC) values and P-Values as CSV files, as these two values are the markers for gene expression levels. The first method used was Volcano Plots, which were created using matplotlib and pandas on Jupyter Notebook to identify the top 40 significantly upregulated genes in red (Figure 2). To plot, significant thresholds were calculated for L2FC > 1 and P-Value < 0.05. To have a nicer figure that summarizes the top upregulated genes in each cluster, I created a table and researched using AI to see what type of cells are present in the data (Table 1). The second method involved training a neural network with TensorFlow and Keras for 20 epochs, using a batch size of 32. This was separated into two groups, the first group as Clusters 1,2,3 and the second group as Clusters 5,6,8. I separated the clusters into two groups as they are isolated from each other (Fig. 1). To verify the validity of my data and ensure the data was not overfit, a confusion matrix (Fig. 3) was created for the two groups (Clusters 1,2,3 vs. Clusters 5,6,8), which resulted in a heatmap that demonstrated the classification accuracy of each cluster to verify whether the model learned biological patterns rather than random chance. Then, a Principal Component Analysis (PCA) was performed to project gene data into a 2D space to capture maximum variance. By doing so, I can illustrate any extreme gene outliers in a PCA plot (Fig. 4).

Results

For the volcano plots, all six clusters were plotted to label the top upregulated and downregulated genes. In Cluster 1 (Fig. 2A), the top gene expressed was *TRAC* (T-cell receptor alpha constant), a type of T-cell receptor often found in T-cells. The gene that had the second-highest level of expression was *CTLA4,* and the third gene was *BATF,* which helps with the transcription and activation of T-cells. In Cluster 2 (Fig. 2B), the top three genes that were expressed were *ZAP70, CD8A,* and *CD96*. These genes are also associated with T-cell activation and development. For Cluster 3 (Fig. 2C), the top three genes were *CLEC10A, CSF2RA,* and *FGL2*, and they are genes that help with immune regulation in myeloid cells. Cluster 5 (Fig. 2D) represented *ABI3BP, MGP,* and *FGF7* as the top genes, which help with tissue regulation and can often be found in tumor microenvironments (TME). Cluster 6 (Fig. 2E) had *CRISPLD2, HEYL,* and *GUCY1A2* as the top genes expressed, which seem to be stromal and endothelial cells. Finally, Cluster 8 (Fig. 2F) illustrates the genes *PGR, LONRF2,* and *TMEM120B* as the top three gene expressions for this cluster and could possibly be glandular cells. When training the neural network for Clusters 1, 2, and 3, the confusion matrix illustrated in Figure 3A highlights that Cluster 3 had the most accurate results, as it had the fewest misclassifications (1499 instances) compared to Cluster 1 (1412 instances) and Cluster 2 (1312 instances). On the other hand, Figure 4B had Cluster 8 with the most correct classifications (1943 instances), with Cluster 6 trailing (1317 instances) and Cluster 5 being the most inaccurate (783 instances). When calculating the accuracy by adding up all of the correct classifications and dividing them by the total number of trials, Figure 4A's accuracy rounds out to about 40.29%, and Figure 4B's accuracy is about 38.57%. To further validate this data, PCA plots were created for each group, and a combined one was created as well. The variance for each plot was calculated and seemed to be low, with 61.90% (Fig. 4A), 61.90% (Fig. 4B), and the combined one with 58% (Fig. 4C). However, the issue with all of the plots is that there are no isolated clusters that seem to differ from each other. To validate my data, I used hand calculations and computational calculations inside my notebook.

Discussion

This project analyzed the 10x Genomics cervical cancer data through single-cell RNA sequencing to determine the gene expression patterns across six different cell clusters. The results of the many identified genes in the clusters represent immune and stromal cells rather than epithelial cells, like squamous and glandular cells. Cluster 1 showed strong expressions of TRAC, CTLA4, and BATF, which are typically associated with T-cell activation and signaling immune pathogens. Similarly, Cluster 2 included genes that indicate a group of $CD8^{+}$ cytotoxic T cells, which are the main drivers in destroying and attacking possible cancerous cells to contribute to the anti-tumor responses. Furthermore, Cluster 3 included genes that suggest myeloid cells are involved in identifying foreign pathogens within the immune system and blocking further malignancy. These findings are consistent with the patterns of cervical cancer and HPV infections. On the other hand, other clusters seem to illustrate tissue cells in the TME. For example, Cluster 5's genes, *FGF7* and *MGP,* are associated with fibroblast activity and remodeling tissue. Then, Cluster 6's genes, such as *HEYL* and GUCY1A2, showed a strong connection to endothelial and stromal cells, which are known to work together to maintain tissue homeostasis. Thus, their upregulation indicates that the TME is corrupted and could possibly promote tumor growth. In contrast, Cluster 8 expressed a type of glandular cell, *PGR,* which is a tumor suppressor, but its upregulation suggests that the possible treatments for the patient were getting reduced cancer progression, as it is typically known to be a downregulated gene in CC. The neural network and PCA analyses showed that there was limited separation of the clusters, which I hypothesize to be due to the clusters' closeness in the UMAP (Fig. 1). Their proximity indicates that the cell populations have a lot of overlap in gene expression levels. It is also interesting to note that Figures 4A and 4B had identical variance, which I am not sure if it is a computational mistake or a coincidence. However, looking at the confusion matrix and their accuracies, it is possible that since they are so similar, their variance is also similar. Ultimately, these results highlight the intricacy of a cervical cancer's TME, as many of the upregulated gene expression levels were in immune cells, which could be used for further research in treatments for tumor cancers.

AI Statement:

I mainly used AI to research the upregulated genes I found and what types of cells they are. I found the article I used from Google Scholar and mainly used its background information to support my paper.

Figures

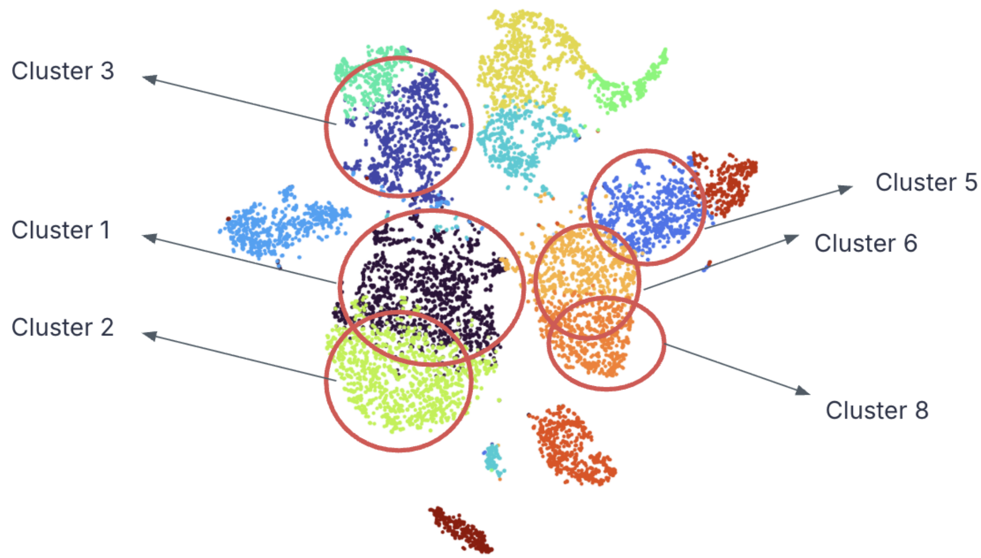

Figure 1. UMAP Projection of the dataset that was taken from the CLOUPE Browser. There are 14 clusters represented within this one dataset, but I chose Clusters 1,2,3,5,6,8, represented by the red circles, to analyze as they are the biggest and separated into two regions.

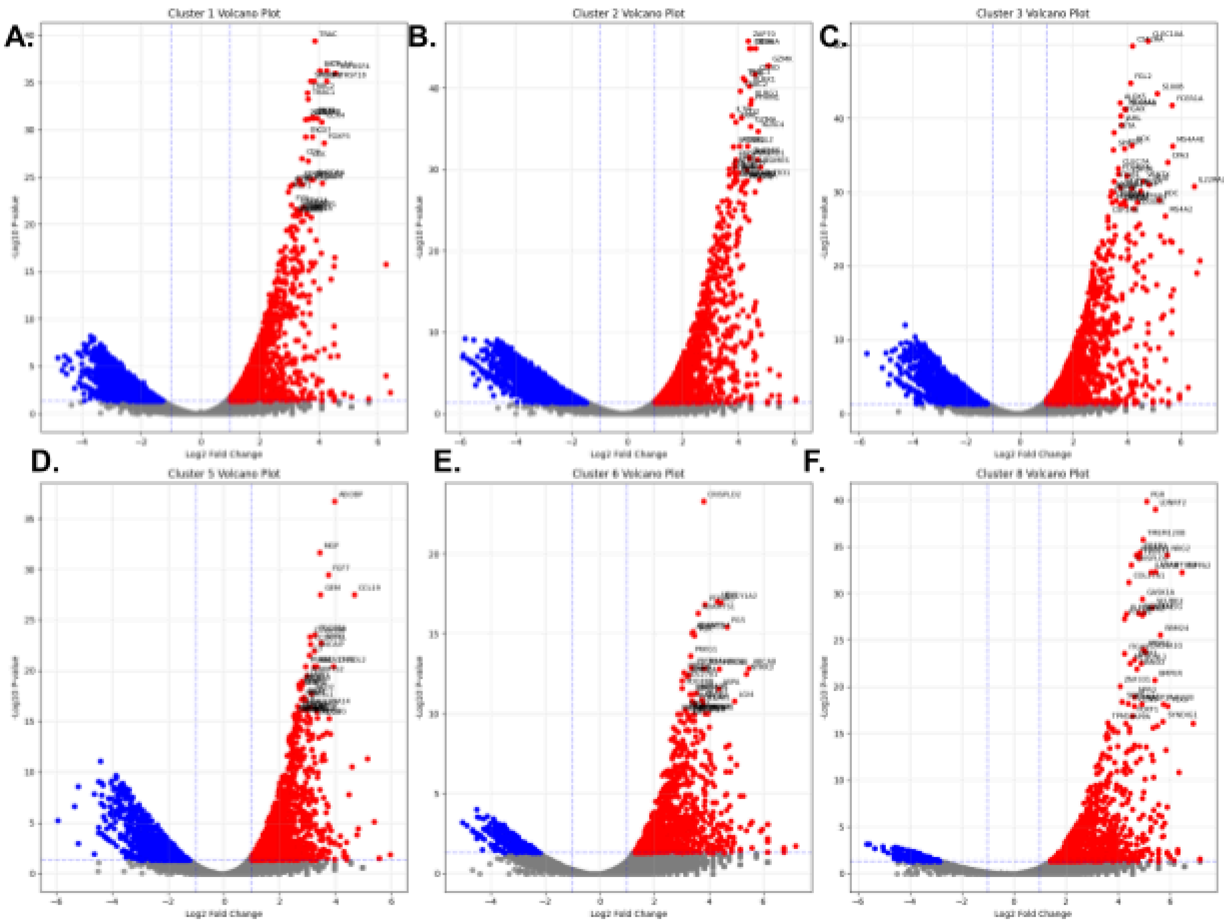

Figure 2. Volcano Plot of Differential Gene Expressions in the Six Clusters. The red points represent the upregulated genes, where the top 20 significant genes are labeled for each cluster, labeled at the top of each plot. The downregulated genes are represented by the blue points. The x-axis represents the Log2FoldChange > 1, while the y-axis represents the $-Log_{10}$ P-Value.

Table 1. Table of Upregulated Genes in Each Cluster. From each cluster, I took the top three upregulated genes seen in the volcano plots and CLOUPE Browser to see if I could identify which cell type they belong to and create conclusions.

Cluster	Top 1 Gene	Top 2 Gene	Top 3 Gene	Type of Cell
Cluster 1	TRAC	CTLA4	BATF	Activated T Cells
Cluster 2	ZAP70	CD8A	CD96	$CD8^{+}$ cytotoxic T cells
Cluster 3	CLEC10A	CSF2RA	FGL2	Dendritic/myeloid cells
Cluster 5	ABI3BP	MGP	FGF7	Cancer-associated fibroblasts
Cluster 6	CRISPLD2	HEYL	GUCY1A2	Stromal/endothelial cells
Cluster 8	PGR	LONFR2	TMEM120B	Hormone-responsive epithelial cells

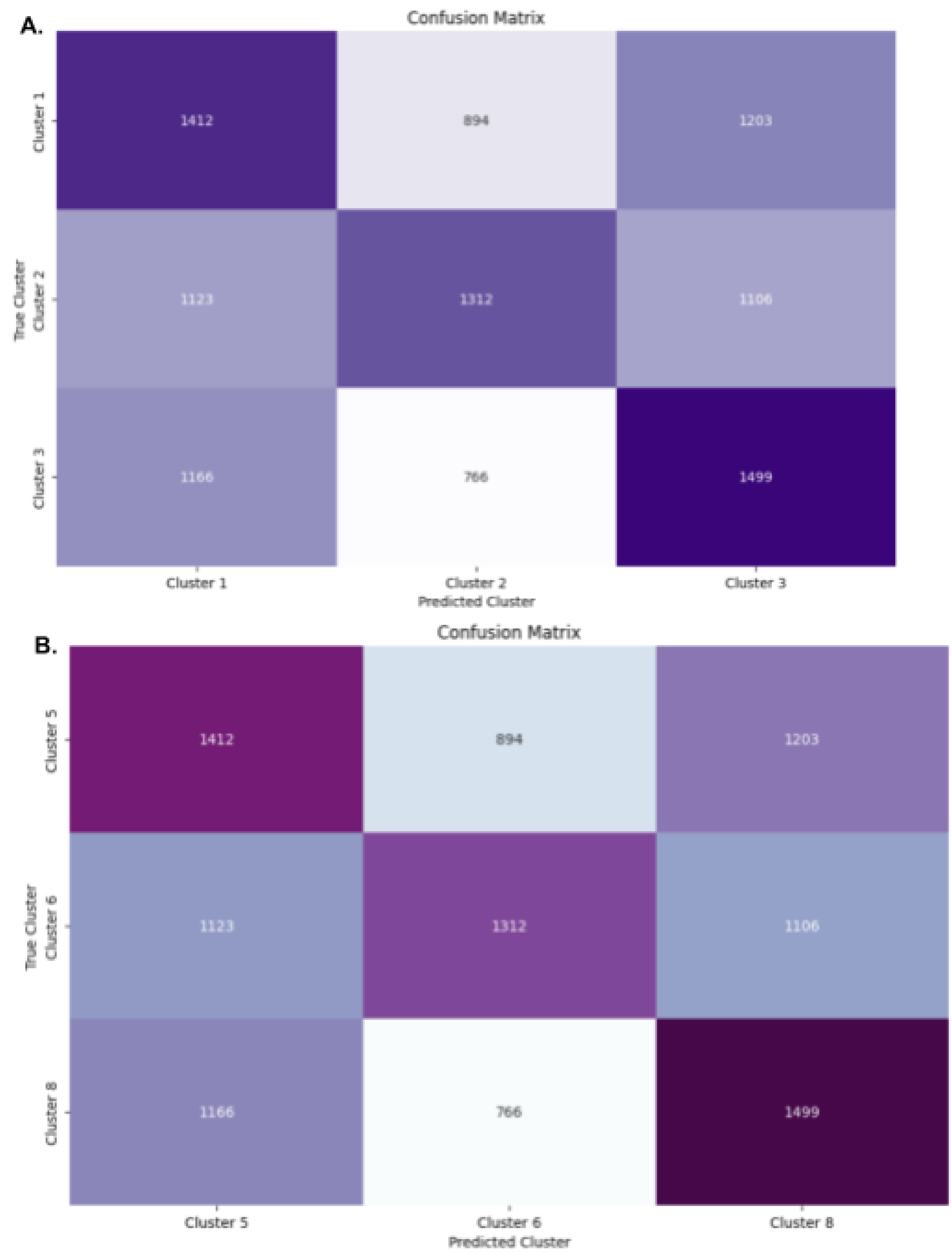

Figure 3. Confusion Matrices of Clusters 1,2,3 and Clusters 5,6,8 Showing that Accuracy is Low. These confusion matrices are split into two groups of clusters. **(A)** Confusion matrix demonstrating the mapping between the true clusters (Clusters 1, 2, 3) and the predicted clusters (Clusters 1, 2, 3). **(B)** An identical distribution was mapped to the other clusters (Clusters 5, 6, 8) to verify the neural network corresponding to each matrix.

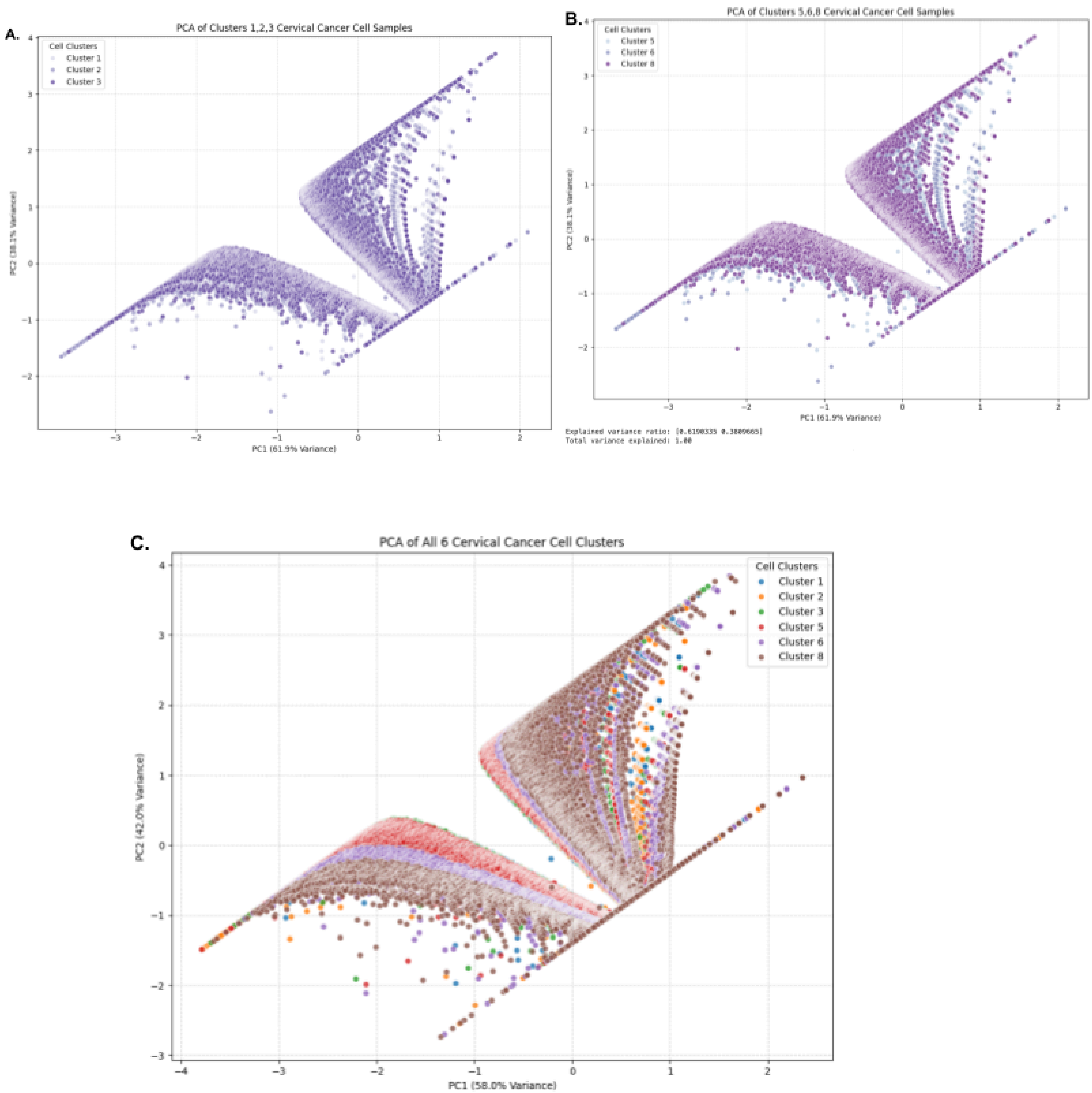

Figure 4. Principal Component Analysis (PCA) Plots Show High Overlap of All Clusters. All three subfigures show principal component projection of high-dimensional single-cell cervical cancer data onto the first two principal components (PC1 and PC2). **(A)** Distribution and overlapping spatial visualization of Clusters 1, 2, and 3. **(B)** Distribution and overlapping spatial

visualization of Clusters 5, 6, and 8, illustrating a high spatial overlap of all three clusters. **(C)** Combined projection of all six cervical cancer cell clusters, which also demonstrates there is a significant overlap and shared variance values throughout the clusters.

References

Human Cervical Cancer FFPE Single Cell Gene Expression flex(Next Gem). (n.d.). 10x Genomics. Retrieved May 5, 2026, from https://www.10xgenomics.com/datasets/17k-human-cervical-cancer-scFFPE

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